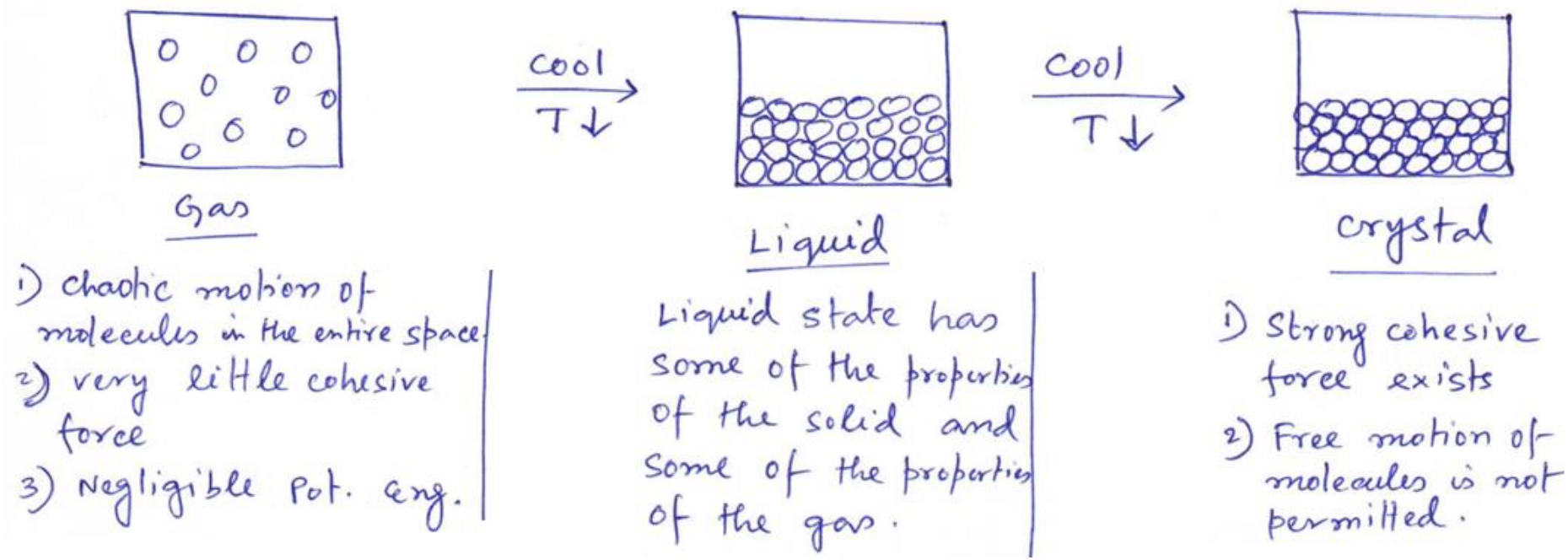
The Liquid State

Typical Physical Properties of Liquids:

(1) Liquid Crystal Behavior
(2) Vapor Pressure
(3) Surface Tension
(4) Viscosity



Liquid Crystal:

For some crystalline substances, it is observed that during heating they form a milky, translucent, anisotropic substances before the formation of clear liquid. This milky substance is called liquid crystal.

So, Crystal → Liquid Crystal → Liquid

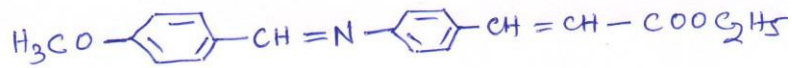
Liquid Crystal

Definition of Liquid Crystal:

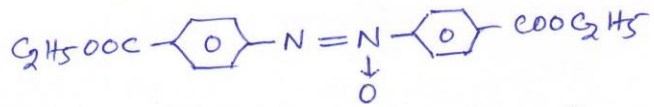
In some substances the tendency toward an ordered arrangement is so great that the crystalline form does not melt directly to a liquid phase at all, but first passes through an intermediate stage (mesomorphic state), which at higher temp. undergoes a transition to a liquid state. This intermediate state has been called liquid crystal since their properties are intermediate between liquid and solid.

Examples:

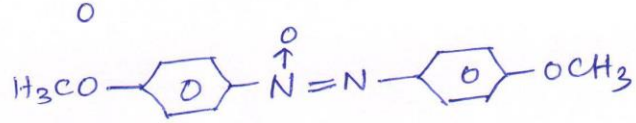
① Ethyl anisal-p-amino cinnamate



② Ethyl p-azoxy benzoate



③ p-azoxyanisole :



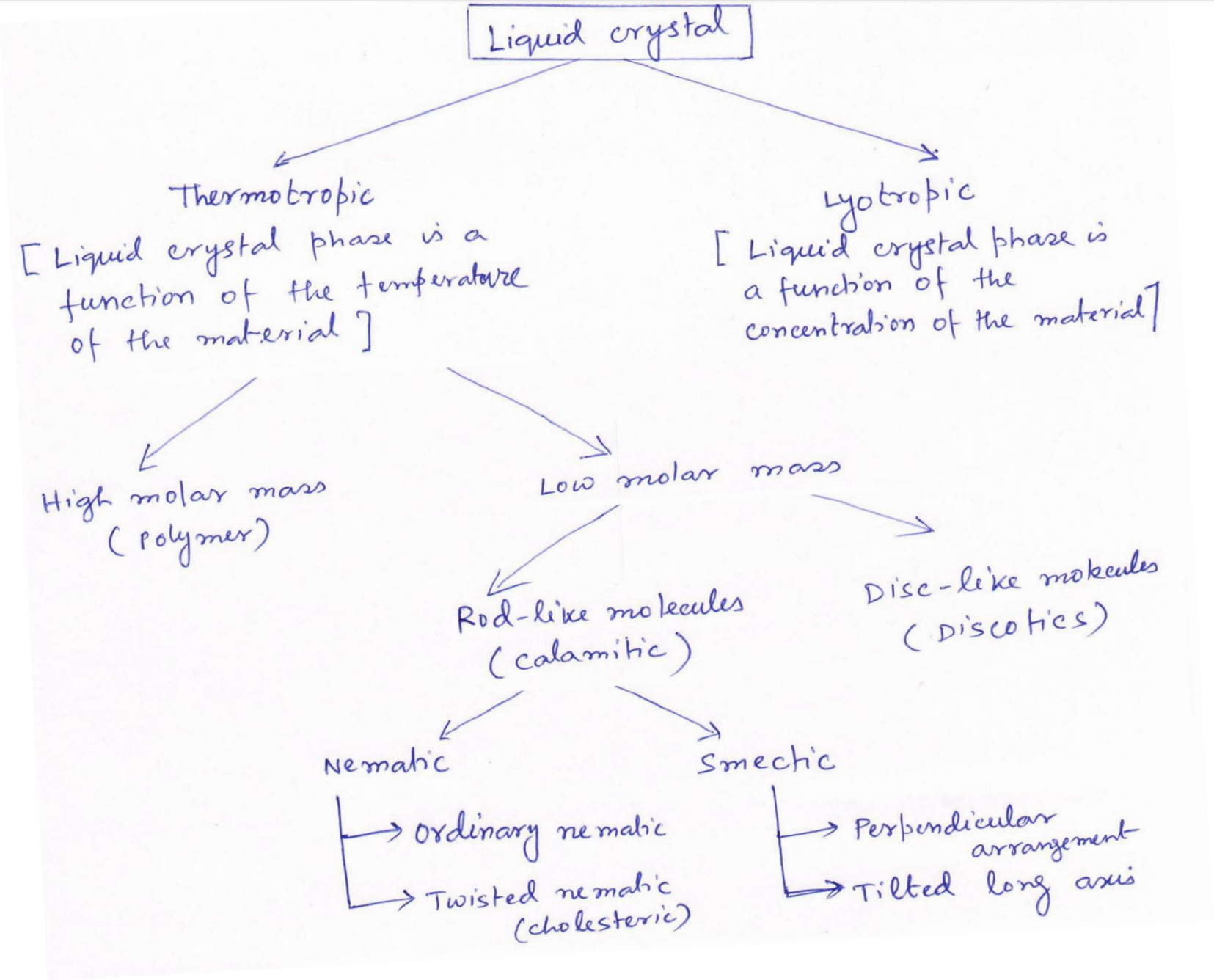
④ Few polymers, substituted cholesteryl ester, and few soaps also form liquid crystals.

Way of formation of Liquid Crystal:

Crystalline solid $\xrightarrow{\Delta}$ Liquid

- ✓ **Kinetic Eng., hence thermal agitation of molecules increases**
- ✓ **It totally disrupts the binding attraction of molecular units of the crystal**
- ✓ **When thermal agitation overcomes the force of cohesion, liquid is formed from solid**
- ✓ **In case of some crystalline solid, the tendency of ordered arrangement is so great that thermal agitation is not sufficient to break the lattice arrangement to some extent**
- ✓ **A milky, translucent substance (liquid crystal) is formed before the formation of liquid**
- ✓ **Also known as mesomorphic state/phase or mesophase**

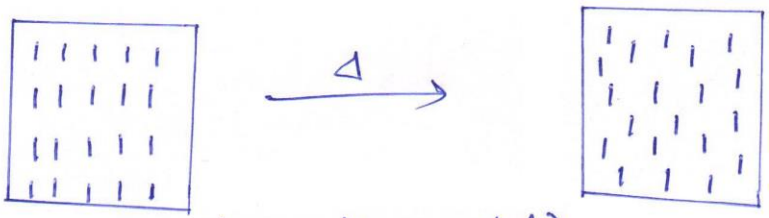
Types of Liquid Crystal



Nematic and Smectic Liquid Crystal

Nematic Liquid Crystal:-

- ✓ 'Nematic' comes from the Greek word 'Nematon', means 'thread-like'
- ✓ The thread like structure of liquid crystal is known as nematic liquid crystal
- ✓ Can be observed by polarized light, using polarized optical microscopy instrument
- ✓ This type of liquid crystal is orientationally ordered, i.e., when stress is applied to it, the planar structure is lost but the orientation is preserved
- ✓ Usually rod-like molecules having length 20 Å forms nematic crystal
- ✓ Examples: (1) 4-Cyano-4'-pentylbiphenyl
  (2) *p*-Azoxy anisole
  (3) *p*-Methoxy cinnamic acid, etc.



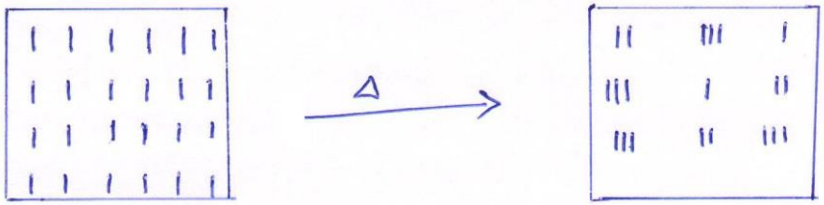
(Nematic crystal)

- ✓ This crystal is widely used in LCD

Smectic Liquid Crystal:-

- ✓ It has both translational and orientational order
- ✓ Translation in 1-D and orientational order at layer
- ✓ When a stress is applied, one plane glides over another
- ✓ Can be identified by polarized light, and X-ray
- ✓ Examples: (1) Ammonium oleate
  (2) Ethyl-*p*-azoxy benzoate
  (3) Ethyl-*p*-azoxy cinnamate, etc.

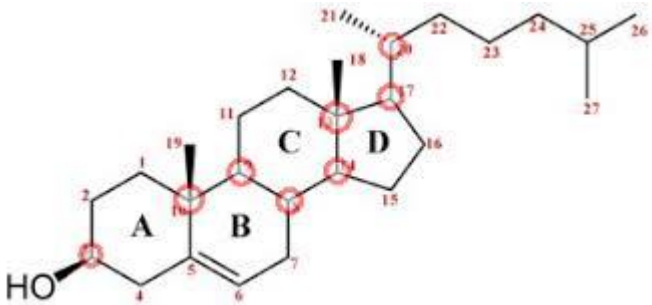
(Smectic crystal)

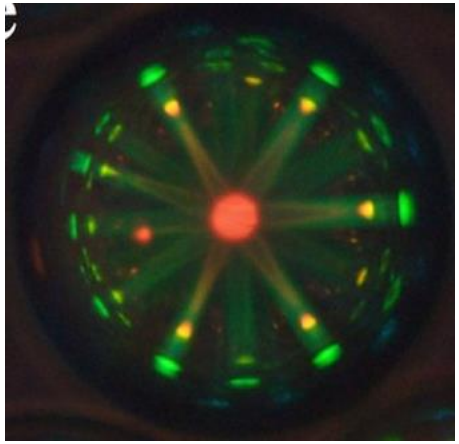


(Different planes or sheets glide over one another)

Cholesteric Phase:-

- ✓ Some liquid crystals, such as cholesteryl esters show in addition to their nematic behavior some color effect under polarized light.
- ✓ They are called ‘cholesteryl phase’ or ‘twist nematic’.
- ✓ Here, incorporation of chiral group gives helical twist to the orientation of the long axis of the molecule.





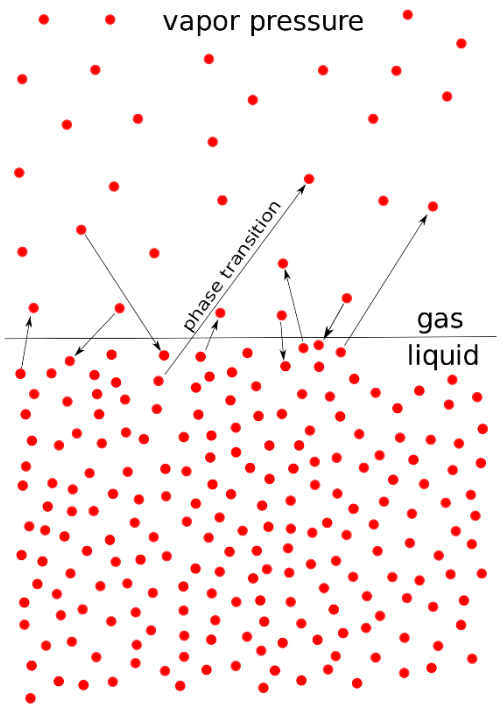
Use of Liquid Crystals:-

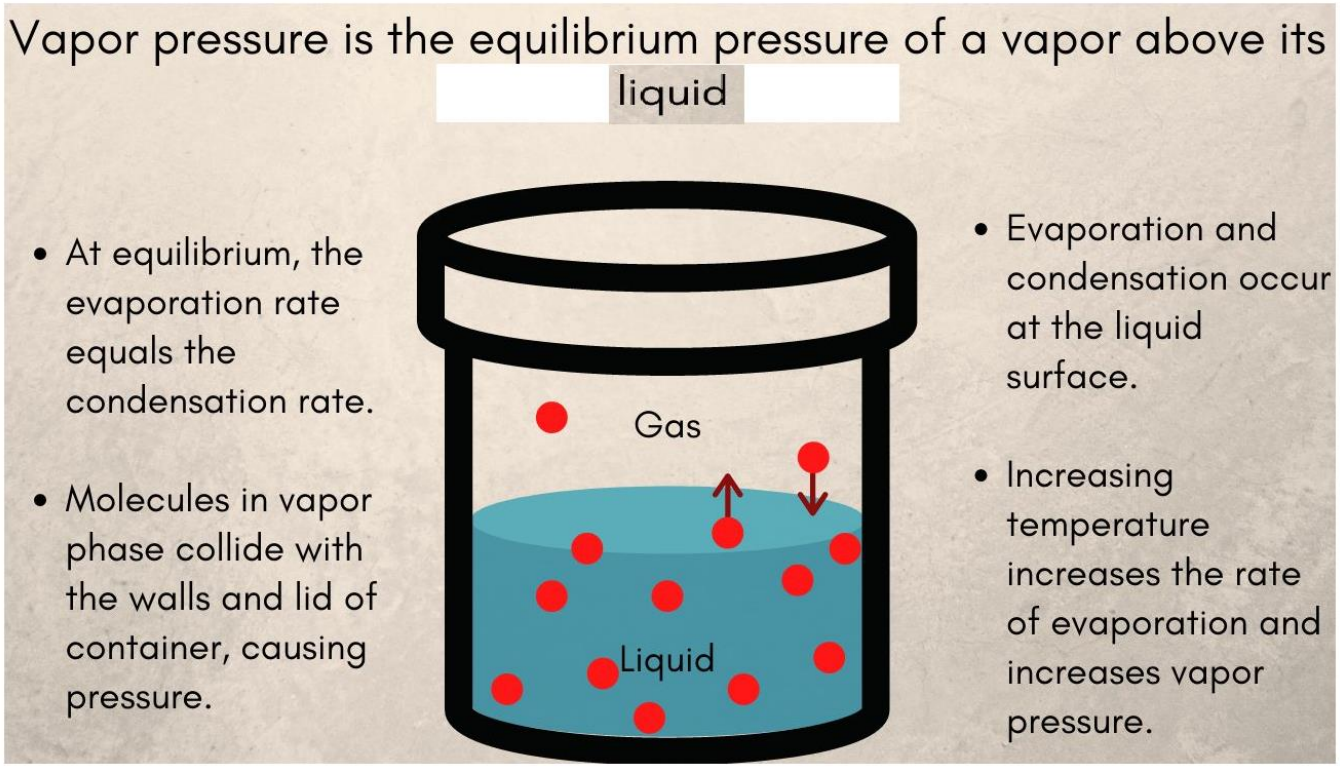
- ✓ Used in LCD: Display of calculator, mobile, TV, clock, etc.
- ✓ It is used as temperature sensors.
- ✓ It is used in automatic gate and window.
- ✓ It is used for the treatment of many skin diseases.
- ✓ This study helps to understand the movement of plasma-like substances in living tissues as in muscles, nerve-sheaths, etc.

Vapour Pressure

Definition:-

- ✓ **Vapour pressure is defined as the pressure of the vapour in equilibrium with its liquid.**





Effect of Temperature on Vapour Pressure

According to clausius-clapeyron Equation: $\frac{dP}{dT} = \frac{\Delta H}{T \Delta V}$

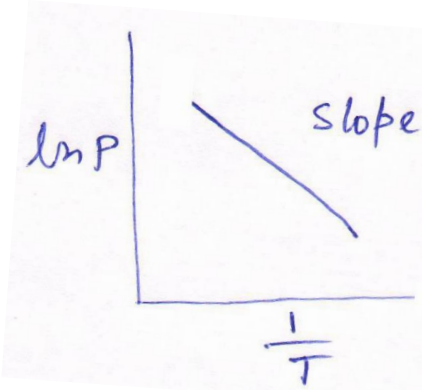
For vapourization of 1 mole of a liquid,

$$\frac{dP}{dT} = \frac{\Delta \bar{H}_{vap}}{T(\bar{V}_g - V_l)} \simeq \frac{\Delta \bar{H}_{vap}}{T \bar{V}_g} ; \left[\begin{array}{l} \bar{V}_g \gg \bar{V}_l \\ \therefore \bar{V}_g - \bar{V}_l \simeq \bar{V}_g \end{array} \right]$$

Thus, $\frac{dP}{dT} = \frac{\Delta \bar{H}_{vap}}{T \bar{V}_g} = \frac{\Delta \bar{H}_{vap}}{T \cdot \frac{RT}{P}}$ (Assuming ideal behaviour)

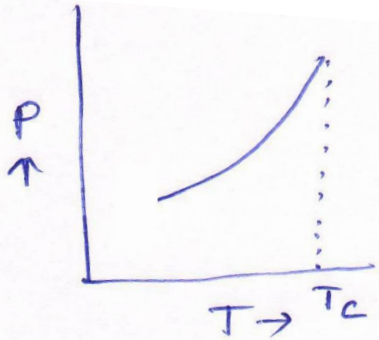
$$\Rightarrow \frac{dP}{P} = \frac{\Delta \bar{H}_{vap}}{RT^2} dT$$

Integrating, $\ln P = -\frac{\Delta \bar{H}_{vap}}{RT} + z$; [z = Integration constant]



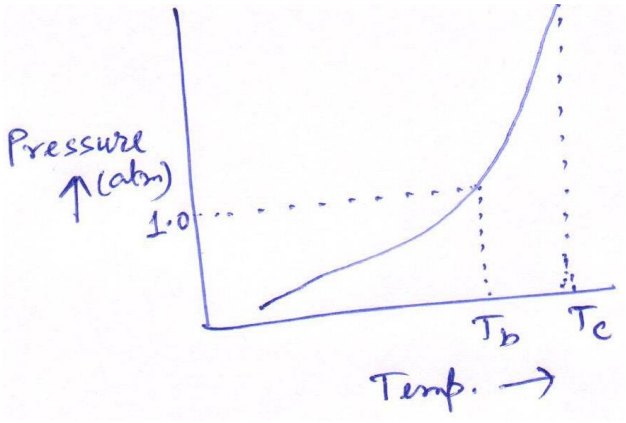
Slope $= \frac{\Delta \bar{H}_{vap}}{R}$

Thus, from this plot $\Delta \bar{H}_{vap}$ can be calculated.



- ✓ **With the rise in temp., the vapour pressure increases.**
- ✓ **The temp. at which the vapour pressure is equal to one atmosphere is the normal boiling point of the liquid.**

Effect of Temperature on Vapour Pressure



- ✓ **There is an empirical relation, applicable in most liquids, between the critical temp. (T_c) and the normal boiling point (T_b) of a liquid.**

Generally, $\frac{T_b}{T_c} = 0.67 = \frac{2}{3}$ **[Temp. in absolute scale]**

The ***critical temperature*** of a substance is the temperature at and above which vapor of the substance cannot be liquefied, no matter how much pressure is applied.

Vapour Pressure and External Pressure:-

- ✓ **We noticed that when a liquid is subjected to higher external pressure, the vapour pressure changes.**
- ✓ **Thermodynamically it can be shown that, at a given temp.,**

$$V_g \, dp = V_l \, dP \; ;$$

p = vapour pressure
P = external pressure
V_g = Molar vol. of the vapour
V_l = Molar vol. of the liquid

Assuming ideal gas law, $\frac{dp}{p} = \frac{V_l}{RT} dP$; $\left[pV_g = RT \Rightarrow V_g = \frac{RT}{p} \right]$

$$\Rightarrow \ln \frac{p_1}{p_2} = \frac{V_l}{RT}(P_1 - P_2)$$

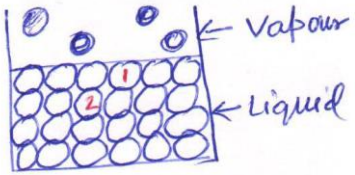
Here, p_1 and p_2 are the vapour pr.s under external pr.s P_1 and P_2. The molar volume of the liquid V_l is regarded as independent of pr.

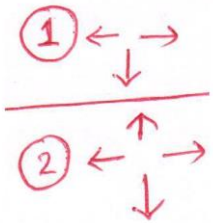
Note: When a liquid is boiling, its vapor pressure is equal to the external pressure. For example, as water boils at sea level, its vapor pressure is 1 atm because the external pressure is also 1 atm.

Please study how experimentally vapour pressure can be determined.

Surface Tension

- ✓ Liquid molecule at the surface feels a resultant inward pull towards the bulk
- ✓ So, liquid surface tends to minimize its area by taking a spherical shape
- ✓ Thus, minimum number of molecules remain at the surface
- ✓ Thus, surface remains under stretched condition or under tension, which is called surface tension or interfacial tension





Definition: Surface tension is defined as the force acting normally on an imaginary line of unit length on the surface.

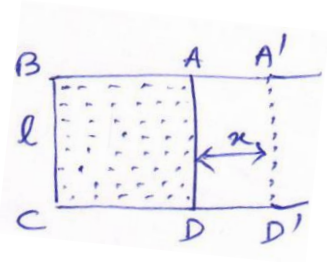
$$\text{surface tension} = \frac{\text{Force}}{\text{Length}}$$

unit of surface tension:-

CGS : dyn/cm

SI : N/m

Surface Energy:-



- ✓ Consider a liquid film ABCD contained in a rectangular wire-frame, where the slide AD is movable
- ✓ Suppose now the film is stretched by moving the boundary AD by x cm to A′D′ position
- ✓ The amount of work required to take sliding partition from AD to A′D′;

$$W = \gamma \times l \times x$$
$$= \gamma \times \Delta A$$
$$\Rightarrow \gamma = \frac{W}{\Delta A}$$

[l = length of AD
γ = Surface tension acting per cm
ΔA = increase in area]

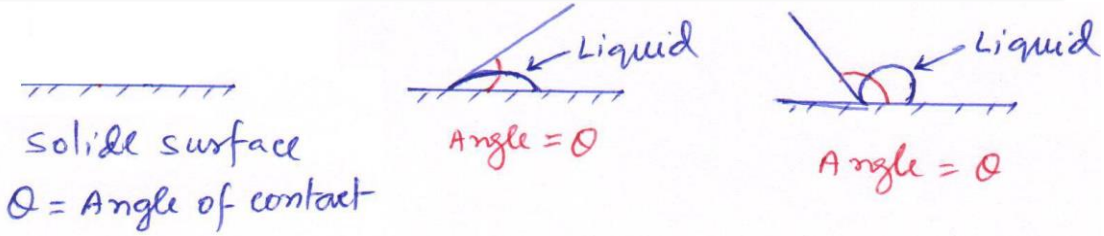
W = F × d F = Force, d = displacement
Unit: N-m

- ✓ ***Alternative definition of surface tension:*** Amount of work required to increase a surface area by unity
- ✓ Surface energy is numerically equal to the work required to build up the surface
- ✓ Unit: dyn/cm and N/m [Consider the dimension of surface tension and surface energy]
- ✓ Since dimensions of surface tension and surface energy are equal, they are equivalent

Angle of Contact

Angle of Contact:

The angle between the tangent to the liquid surface at the point of contact and the solid surface, inside the liquid, for that pair of liquid and solid.



i) when $\theta < 90^\circ$, spreading / wetting occurs

e.g. water on glass surface ; $\theta = 18^\circ$

Parafin on glass ; $\theta = 27^\circ$

Oil on clothes, alcohol on glass, etc.

ii) when $\theta > 90^\circ$, no spreading or wetting occurs

e.g. Hg on glass ; $\theta = 110^\circ$

H_2O on solid parafin ; $\theta = 140^\circ$

H_2O drop on a lotus surface ; $\theta = 148^\circ$

Superhydrophobicity:-

- ✓ **Superhydrophobic surfaces are highly hydrophobic, i.e., extremely difficult to wet**
- ✓ **The contact angles of a water droplet on a superhydrophobic material exceeds 150°**
- ✓ **Many natural surfaces, such as plant leaf, insects, etc. exist around us**